

TAMIL NADU STATE BOARD - STANDARD 12 CHEMISTRY VOLUME 1

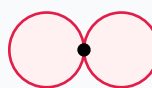
CHAPTER 5: COORDINATION CHEMISTRY

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Chemistry Volume 1
Chapter Name:	Coordination Chemistry

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

p orbital (px)

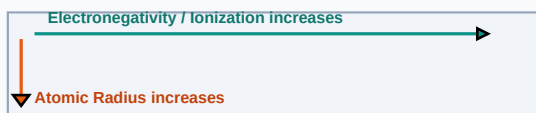
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Ligand	An ion or molecule bound to a central metal atom in a coordination complex by donating electron pairs.
Coordination Number	The number of donor atoms of ligands directly attached to the central metal ion in a complex.
Chelation	The formation of two or more separate coordinate bonds between a polydentate ligand and a single central metal.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

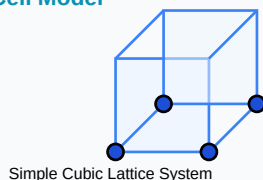
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Effective Atomic Number: $EAN = Z - (\text{Oxidation State}) + 2 * (\text{Coordination Number})$
- IUPAC Prefix: di, tri, tetra | Polydentate: bis, tris, tetrakis

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

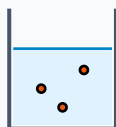
Example 1: Calculate EAN of central metal in $[\text{Fe}(\text{CN})_6]^{4-}$ (Z of Fe = 26).

Step-by-step Solution:

Oxidation state of Fe is +2. Coordination number is 6.

$\text{EAN} = 26 - 2 + 2(6) = 24 + 12 = 36$ (achieves Krypton noble configuration).

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Apply Valence Bond Theory to predict hybridization, geometry, and magnetic nature of $[\text{CoF}_6]^{3-}$.

Analytical Solution Strategy:

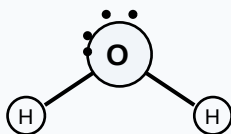
Co is +3 state ($[\text{Ar}] 3d^6$). F^- is a weak field ligand, cannot pair electrons.

Unpaired electrons: 4. Hybridization involves outer orbitals: sp^3d^2 .

Geometry: Octahedral.

Nature: High spin paramagnetic complex.

Lewis Dot Structure (H_2O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

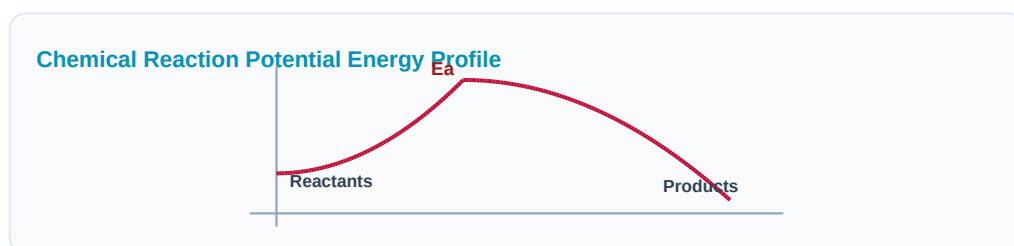
Q1 (1-Mark): State Werner's theory postulates.

Q2 (2-Mark): Write IUPAC name for: (1) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$, (2) $\text{K}_4[\text{Fe}(\text{CN})_6]$.

Q3 (2-Mark): Differentiate between homoleptic and heteroleptic complexes.

Q4 (5-Mark): Explain linkage isomerism and coordination isomerism with examples.

Q5 (5-Mark): What is crystal field splitting energy (CFSE)?



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Medicine

EDTA chelates heavy metal ions like lead to treat clinical lead poisoning cases.

Application Case: Photography

Sodium thiosulfate washes away unexposed silver bromide complexes during film development.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

